

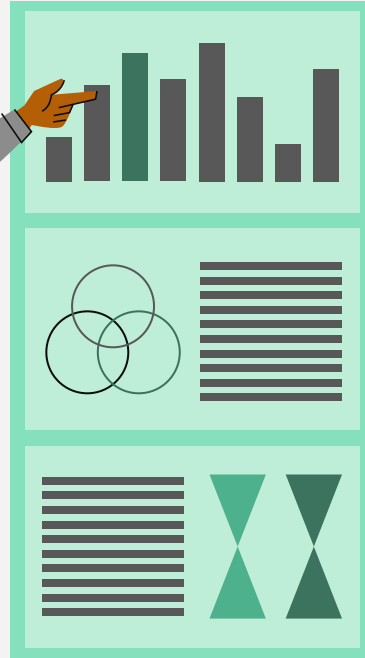
Conversion Prediction for Digital Marketing Campaigns

Team 9

Amisha Kelkar, Manan Goel,
Pratik Mahajan, Yashna Meher



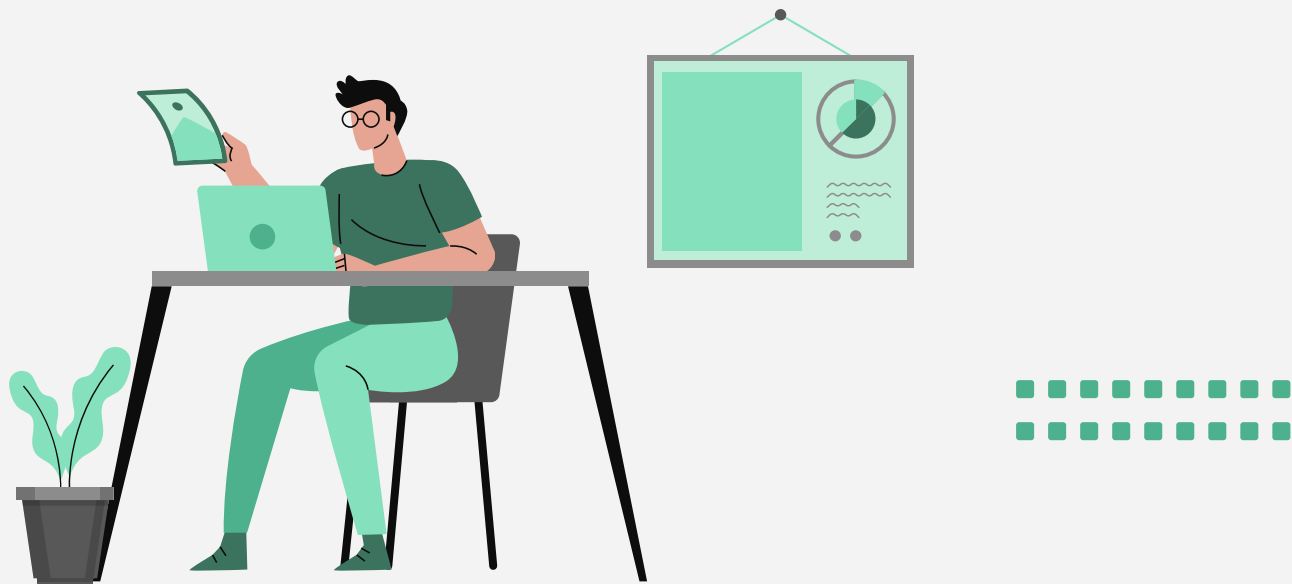
Agenda



- 01** Introduction
- 02** Our Data
- 03** Exploratory Analysis
- 04** Methods and Results
- 05** Challenges
- 06** Conclusion



01 Introduction





Problem Statement



Improving customer conversion rates

Predict which customers are more likely to convert



Optimize Campaigns

Tailor marketing campaigns to the customers who are most likely to engage, leading to better use of their advertising budget.



Improve Engagement

Increase customer engagement by focusing on the right channels and content that resonate with the target audience.



Enhance ROI

Drive more sales and improve ROI by identifying customers who are most likely to take action.





Stakeholders

Marketing Team

To refine campaign strategies by predicting which customers are likely to convert.

Sales Team

To focus efforts on high-potential leads and improve sales conversion rates.

Product Manager

To assess the effectiveness of marketing channels and tools.

Executives

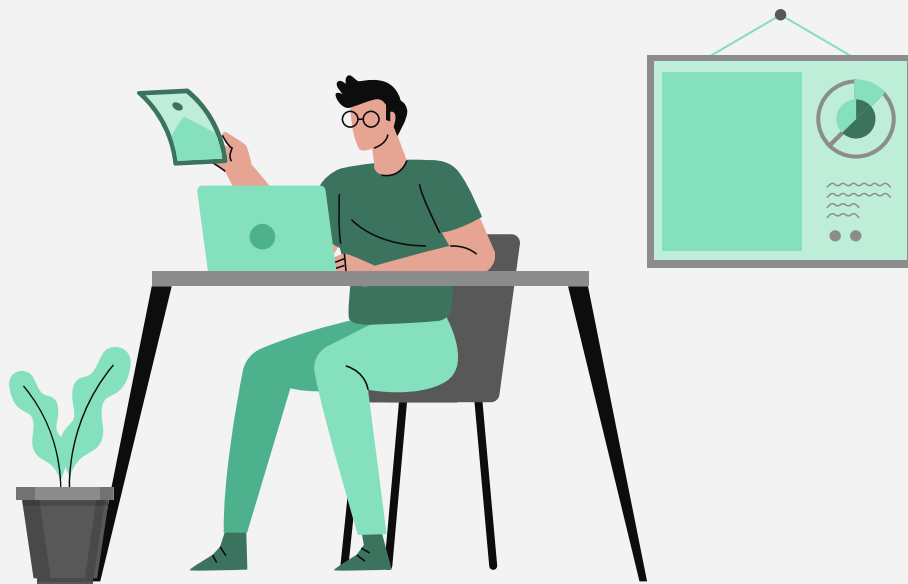
To optimize budget allocation across campaigns and channels.

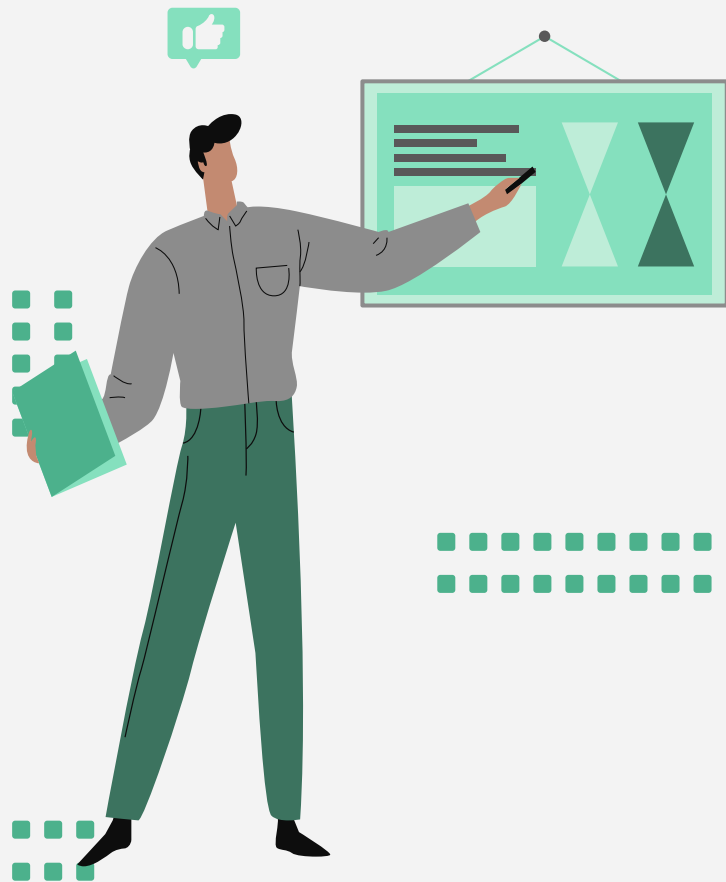
Data Analysts

To derive insights from customer behavior and enhance predictive models.



02 Our Data





Mr. R. E. Kharoua

Synthetic Digital Marketing Dataset from Kaggle
generated for educational purpose

16,000 rows

Distinct Customers

18 Features

And 1 Target Variable to use in ML models



Dataset Details

Demographic

- CustomerID
- Age
- Gender
- Income

Marketing-specific

- CampaignChannel
- CampaignType
- AdSpend
- ClickThroughRate
- ConversionRate
- AdvertisingPlatform
- AdvertisingTool

Customer Engagement

- WebsiteVisits
- PagesPerVisit
- TimeOnSite
- SocialShares
- EmailOpens
- EmailClicks

Historical Data

- PreviousPurchases
- LoyaltyPoints

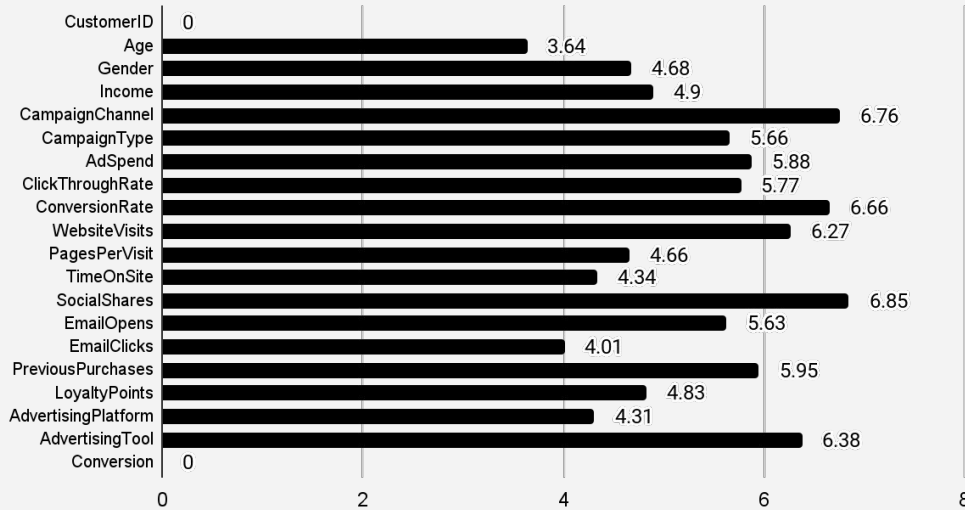
Target Variable

- Conversion
- 

03 Exploratory Analysis



Percentage Null Values



Click-Through Rate:
14.6%

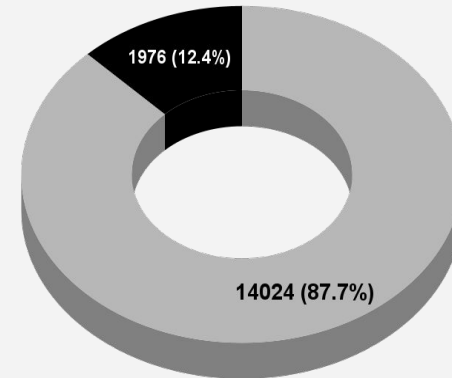
Cost per Acquisition:
\$5373.68



Age Range:
18 – 69 (Avg 43)

Income Range:
\$20k – \$150k

Conversion Value Count



04 Methods and Results



Methodology



Data Analysis and Preprocessing

Handled missing values, applied custom pipelines, and visualized key trends.

Feature Selection

Applied forward and backward feature selection to identify key predictors.

Ensemble Method

Stacked models using ensemble learning to achieve the highest performance.

Model Evaluation

Compared balanced accuracy of models (XGBoost, Decision Tree, Logistic Regression, Random Forest) addressing class imbalance effectively.

Hyperparameter Tuning

Used grid and random search to optimize model parameters.

	Logistic Regression Backward Grid	Random Forest Backward Grid	Decision Tree Backward Random	XGBoost Backward Random
	65.70% → 68.14%	63.92% → 68.13%	69.31% → 70.83%	67.16% → 68.69%

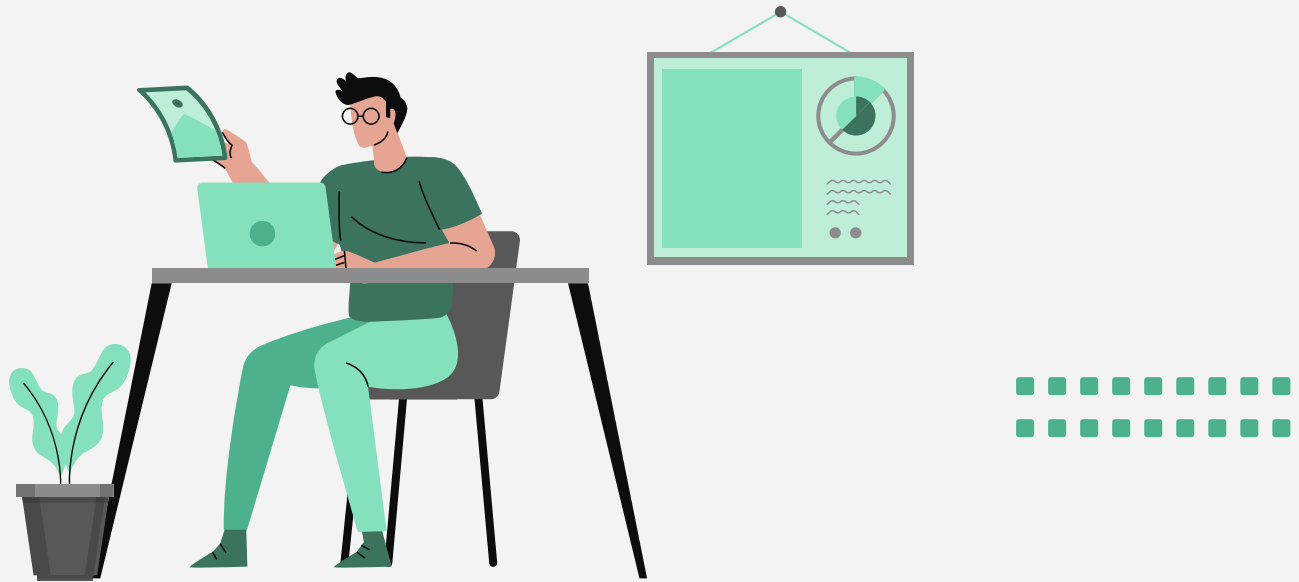


80.16%

Accuracy after using Stacking
ensemble method



05 Challenges





Synthetic Data Characteristics

Limited correlations and trends tested model robustness and preprocessing techniques.

Data Augmentation

Synthetic data scaled and validated to match the Kaggle dataset effectively.

Class Imbalance

Required techniques like class weighting to address imbalance in conversion rates.

Computational Complexity

Feature selection and cross-validation significantly increased processing time.

Model Tuning Challenges

Extensive iterations needed to balance performance and interpretability effectively.

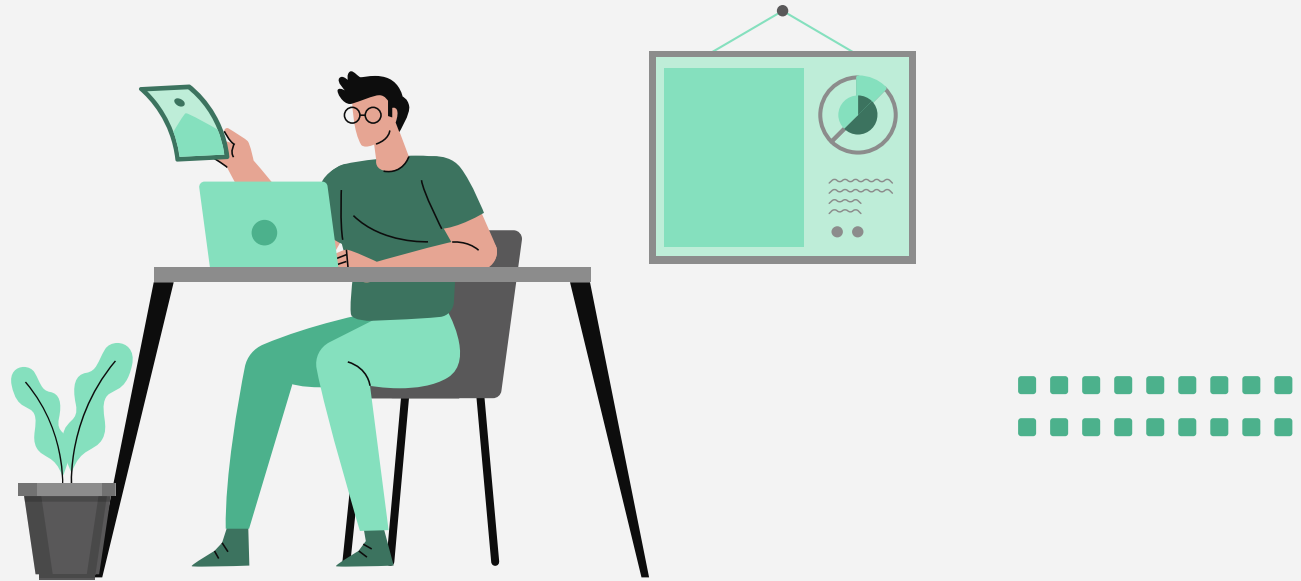


Class Imbalance Solutions

SMOTE-NC caused overfitting; class weighting proved the best approach.



06 Conclusion



Model Evaluation and Results



Decision Tree Classifier

Achieved high balanced accuracy of 78.30%.
Strong initial performer among classifiers.

Stacked Model

Balanced accuracy improved to 80.16%.
Combines strengths of multiple models
effectively.

Preprocessing and Tuning

Critical for optimizing model performance.
Enables consistent improvements across
classifiers.



Learning Outcomes



Balanced Accuracy

Robust metric for imbalanced datasets.
Improves reliability of evaluations.

Synthetic Data Handling

Requires preprocessing and feature engineering.
Essential for extracting meaningful trends.

Ensemble Learning

Effectively mitigates weaknesses of individual models.
Boosts overall predictive performance.

Practical Implications

Predictive models guide marketing resource allocation

Real-world datasets improve model interpretability

Data-driven decisions enhance marketing efficiency



Thank You



[Colab Notebook Link](#)

